Why Some People Hear Music That Isn't Really Playing

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Almost everyone has had a song stuck in their head. But some people hear music that sounds just as real as a song on the radio, even when no music is playing. Scientists call these musical **hallucinations**.

A hallucination is when the brain senses something that is not really there. Musical **hallucinations** are sometimes linked to **hearing loss**. One idea is that when the part of the brain that handles sound gets less information from the ears, it starts pulling music from a person's memory instead.

A new study in the **Journal of Neurology** looked more closely at this condition. Researchers from the Netherlands and the United Kingdom studied 80 people who hear music that is not there. Most were older adults, with an average age of 65. The **participants** answered questions about their experiences and took hearing tests and brain scans.

The scientists noticed some interesting patterns. People who had **hearing loss** that was worse in one ear usually heard the music as if it came from outside their heads. Most others heard it inside their minds.

Most of the songs were older music, not new hits. Religious songs were the most common type, reported 18 percent of the time. Earlier research also found that these songs were often holiday or children's songs. This suggests the brain may be reaching into **long-term memory** for tunes people learned when they were young.

For many **participants**, the music was hard to control and bothersome. Only about one in five could do anything to change it, such as playing real music to cover it up.

Treatment depends on the cause, and explaining the science to patients can help. The goal is to remove the shame some people feel about the condition, so they are more comfortable asking for help.

Word Treasure

hallucinations -- Experiences of seeing, hearing or feeling something that is not really there.

hearing loss -- When a person cannot hear sounds as well as before.

participants -- The people who take part in a study.

long-term memory -- The part of memory that stores information for years, like songs learned as a child.

Journal of Neurology -- A science magazine where researchers publish studies about the brain and nerves.

Background you need

The brain does not just receive sounds from the ears; it actively works to make sense of them and fill in gaps.

A part of the brain handles hearing, and when it gets less information from the ears, it may start producing sounds on its own.

An ordinary 'earworm' is a song stuck in your head that you know is not real, while a musical hallucination sounds like real music playing.

Scientists study people with this condition using questionnaires, hearing tests and brain scans to understand what causes it.

Break it down

LEAD: Some people hear music that sounds real even when nothing is playing
KEY IDEA: These are musical hallucinations, sometimes linked to hearing loss
+- THEORY: When the brain's sound area gets less input from the ears, it pulls music from memory
STUDY: Researchers from the Netherlands and the United Kingdom studied 80 people, average age 65
+- METHOD: Questions, hearing tests and brain scans
+- FINDINGS: One-sided hearing loss made music seem outside the head; most songs were old, religious songs most common
EVIDENCE: Earlier research found holiday and children's songs, pointing to long-term memory
IMPACT: Only about one in five could control the music; explaining the science may help people seek help

Why it matters

Understanding why the brain creates music that is not there can reduce shame around the condition and help people get the right support.

Test what you remember

1. What do scientists call it when people hear music that is not really playing?

(A) Earworms
(B) Echoes
(C) Sound waves
(D) Musical hallucinations

2. How many people did the new study look at?

(A) 80
(B) 8
(C) 18
(D) 800

3. What was the average age of the participants?

(A) 85
(B) 65
(C) 35
(D) 15

4. Where did people with hearing loss worse in one ear usually hear the music coming from?

(A) From their phones
(B) Inside their heads
(C) Outside their heads
(D) Only in dreams

5. Which type of song was reported most often?

(A) Rap songs
(B) New pop hits
(C) Religious songs
(D) Movie theme songs

6. About how many participants could do anything to control the music?

(A) None of them
(B) About one in five
(C) All of them
(D) About half

Answer key (try the quiz first!): 1: D 2: A 3: B 4: C 5: C 6: B

Questions to think about

MY THOUGHTS (at least 20 words):